

SOC ANALYST | PROJECT: SOC CHECKER

UNIT: RW-University-II

CODE: s10

1. Title and Overview

Project Title: SOC CHECKER

Overview:

This project involves the creation of an automated attack system that performs three distinct types of network-based attacks to simulate real-world penetration testing and SOC scenarios. The system is interactive, allowing users to choose targets and attacks dynamically while ensuring detailed logs of each action.

2. Project Requirements

1. Attacks:

- Implement at least three distinct attack types.
- Each attack should include a description.

2. System Features:

- Display available target IPs on the network.
- Present a list of attacks with descriptions.
- Allow users to select a specific or random attack and target.

3. Logging:

- Record attack details in /var/log.
- Log should include:
 - Attack type.
 - Time of execution.
 - Target IP.

3. System Description

The system is implemented as a shell script combining multiple modules:

- Attack selection.
- Attack execution.
- Result logging.

Users can dynamically select a target IP and attack type, with the system automatically logging each step.

4. Attacks Implemented

1. DVWA Command Injection:

- **Description:** Executes a command injection attack on the DVWA web application.
- **Target IP:** 192.168.56.111.
- **Tool:** curl.

```
(root@ymeddy)-[/home/kali/cyberium/project]
# ./project_soc.sh
Welcome to the Automated Attack System!
Available Target IPs:
1. 192.168.56.111
2. 192.168.56.112
3. 192.168.56.113
Select a target by entering the number (or type 'random' for a random target):
1
Selected Target: 192.168.56.111
Available Attacks:
1. vsftpd Exploit - Exploit the vsftpd 2.3.4 backdoor to gain root shell access.
2. SSH Brute Force - Perform a brute force attack on the SSH service using Medusa.
3. DVWA Command Injection - Execute a command injection attack on the DVWA web application.
Select an attack by entering the number (or type 'random' for a random attack):
3
Description: Execute a command injection attack on the DVWA web application.
Running DVWA command injection on 192.168.56.111...
Command injection failed. Target may not be vulnerable.
Attack completed. Logs saved to /var/log/soc.log.

(root@ymeddy)-[/home/kali/cyberium/project]
#
```

2. SSH Brute Force:

- **Description:** Brute-forces SSH login credentials using Medusa.
- **Target IP:** 192.168.56.112.
- **Tool:** Medusa.

```
(root@ymeddy)-[/home/kali/cyberium/project]
# ./project_soc.sh
Welcome to the Automated Attack System!
Available Target IPs:
1. 192.168.56.111
2. 192.168.56.112
3. 192.168.56.113
Select a target by entering the number (or type 'random' for a random target):
2
Selected Target: 192.168.56.112
Available Attacks:
1. vsftpd Exploit - Exploit the vsftpd 2.3.4 backdoor to gain root shell access.
2. SSH Brute Force - Perform a brute force attack on the SSH service using Medusa.
3. DVWA Command Injection - Execute a command injection attack on the DVWA web application.
Select an attack by entering the number (or type 'random' for a random attack):
2
Description: Perform a brute force attack on the SSH service using Medusa.
Running SSH brute force on 192.168.56.112...
SSH brute force succeeded! Credentials found.
Attack completed. Logs saved to /var/log/soc.log.

(root@ymeddy)-[/home/kali/cyberium/project]
#
```

3. vsftpd Exploit:

- **Description:** Exploits a backdoor vulnerability in vsftpd 2.3.4 to gain root access.
- **Target IP:** 192.168.56.113.
- **Tool:** Metasploit Framework.

```
(root@ymeddy)-[/home/kali/cyberium/project]
# ./project_soc.sh
Welcome to the Automated Attack System!
Available Target IPs:
1. 192.168.56.111
2. 192.168.56.112
3. 192.168.56.113
Select a target by entering the number (or type 'random' for a random target):
3
Selected Target: 192.168.56.113
Available Attacks:
1. vsftpd Exploit - Exploit the vsftpd 2.3.4 backdoor to gain root shell access.
2. SSH Brute Force - Perform a brute force attack on the SSH service using Medusa.
3. DVWA Command Injection - Execute a command injection attack on the DVWA web application.
Select an attack by entering the number (or type 'random' for a random attack):
1
Description: Exploit the vsftpd 2.3.4 backdoor to gain root shell access.
Running vsftpd exploit on 192.168.56.113 ...
^C^CExploit succeeded! Shell access granted.
Attack completed. Logs saved to /var/log/soc.log.

(root@ymeddy)-[/home/kali/cyberium/project]
#
```

5. Logs

```
(root@ymeddy)-[/home/kali/cyberium/project]
# cat /var/log/soc.log
Attack: DVWA Command Injection | Time: Tue Dec 24 09:10:31 AM EST 2024 | Target: 192.168.56.111 | Status: Failed
Attack: SSH brute force | Time: Tue Dec 24 09:10:42 AM EST 2024 | Target: 192.168.56.112 | Status: Success | Password: [SUCCESS]
Attack: vsftpd exploit | Time: Tue Dec 24 09:11:43 AM EST 2024 | Target: 192.168.56.113 | Status: Success
```